CodexMap: The Observability Layer for AI-Generated Systems

> Mapping the invisible evolution of AI-driven codebases in real-time.

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1. Executive Summary

As AI agents move from writing simple scripts to managing complex, multi-file systems, developers face a critical **Observability Gap**. **CodexMap** bridges this gap by providing a real-time, high-fidelity "Control Plane" for AI-generated code. It transforms non-deterministic AI outputs into a visual, interactive dependency graph, governed by deterministic architectural "Skills" and heuristic scoring.

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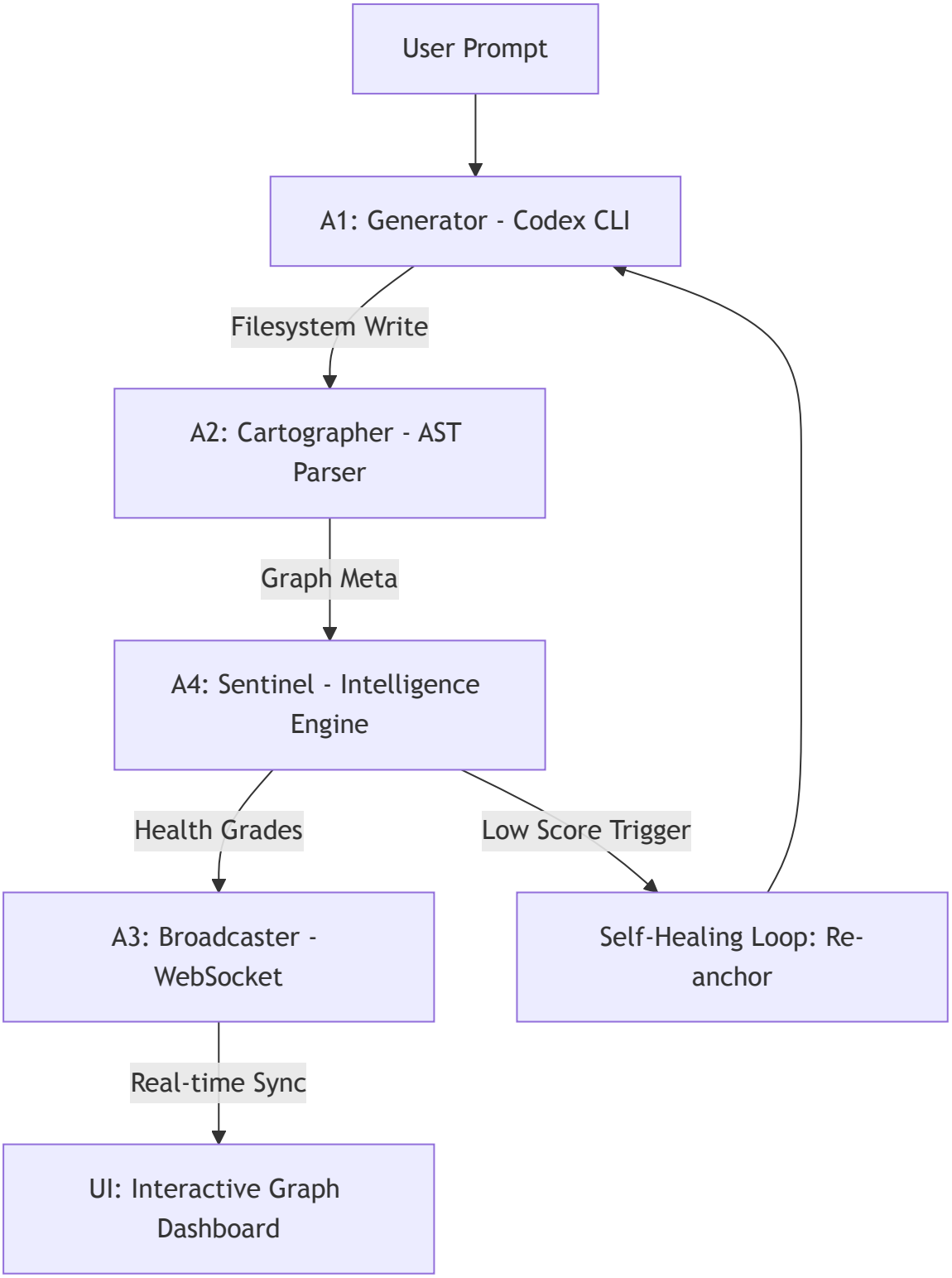
2. System Architecture

CodexMap operates as a multi-agent orchestration layer, ensuring that every line of code is tracked from generation to deployment.



The Orchestration Pipeline:

1. **A1: Generator:** Interacts with the **Codex CLI** in `--approval-mode auto-edit`.
2. **A2: Cartographer:** Uses **Babel AST parsing** and regex fallbacks to decompose files into stable logical nodes (Functions, Classes, Imports).

3. **A3: Broadcaster:** Managed WebSocket layer that diffs and batches graph updates to ensure 60FPS UI performance.
4. **A4: Sentinel:** The primary brain. Runs the **5-Component Hybrid Scoring Engine** to detect context drift.

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3. The Sentinel Intelligence Engine

The heart of CodexMap is the **Sentinel**, which performs non-deterministic evaluation against deterministic project intents. It produces a composite health score (0.0 to 1.0) for every node.

5-Component Scoring Matrix

Component	Logic	Weight	Purpose
S1: Semantic	Cosine Similarity (Vector Embeddings)	20%	Measures baseline intent alignment.
S2: Reasoning	Cross-Encoder & PageIndex Logic	40%	Validates structural and logical coherence.
A: Architectural	DNA Anchoring & Import Graph Fit	20%	Ensures code resides in the correct module domain.
T: Type/Role	AST Validation & Keyword Intensity	20%	Checks for language-specific best practices.
D: Drift Penalty	Anti-Pattern & Subsystem Detection	-30%	Penalizes "hallucinated" features (e.g., unwanted Stripe APIs).

> [!IMPORTANT]
> **Definitive Formula:** `S_final = (0.2*S1) + (0.4*S2) + (0.2*A) + (0.2*T) - (0.3*D)`

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4. Enterprise-Grade Safety & Governance

The Collapse Warning System

CodexMap detects the "point of no return" for AI-generated subsystems before they require manual refactoring.

> [!CAUTION]
> **Architectural Collapse Signals:**
>
> - **Entropy Surge:** Red node density exceeds 40% of the active codebase.
> - **Spaghetti Growth:** Edge count (dependencies) grows >3× the baseline rate in 5 minutes.
> - **Complexity Spike:** Average Cyclomatic Complexity per node doubles from the initial anchor.

Skills & Compliance (`SKILL.md`)

The system is governed by a local `SKILL.md` framework—a set of deterministic "ground rules" for the AI agents:

- **Black-Box Safety:** Agents never parse non-deterministic CLI output; all truth is derived from the **Cartographer's FS Watcher**.
- **Atomic State Management:** Use of temporary buffer writes and rename-swaps prevents data corruption during high-frequency map updates.

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5. Rigorous Evaluation Harness

CodexMap is built on a foundation of measurable reliability. The system includes a comprehensive **Evaluation Suite** that benchmarks the Sentinel engine against both synthetic and ground-truth datasets.

Metrics & Methodology

- **Heal-Efficiency (Mean Score Delta):** The `heal_eval` harness quantifies recovery success. It measures the **Mean Score Delta (MSD)** before and after an automated re-anchoring loop. The system maintains a rigorous success threshold of **$\Delta > 0.15$** for all self-healing operations.
- **Drift Recall & Latency:** Using the `inject_drift` stress-tool, the system purposefully introduces out-of-scope logic (e.g., hallucinated payment gateways) to measure the Sentinel's sensitivity. Key KPIs include:
  - **Detection Recall:** The percentage of synthetic drift events correctly identified as "Red."

  - **Detection Latency:** Time-to-flag (measured in seconds) from injection to UI alert.
- **Semantic Precision Benchmarking:** The `score_accuracy` suite compares heuristic scores against a **Ground Truth Registry**, ensuring the 5-component formula remains aligned with human architectural intent.
- **Pipeline Performance:** Latency testing across the Babel AST parser and WebSocket broadcaster ensures the system scales to codebases exceeding 1,000+ nodes without UX degradation.

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6. Codex Ecosystem Integration

CodexMap leverages the full power of the Codex platform through specialized plugins and skills:

- **Codex CLI App:** Orchestrates the generative lifecycle and manages the re-anchoring process for drifted nodes.
- **PageIndex Plugin:** Builds a deep-index system tree, allowing the Sentinel to "reason" across file boundaries without exceeding context limits.
- **Cross-Encoder Scaling:** Uses dedicated reasoning nodes to perform secondary logic validation, ensuring high-fidelity outputs for production systems.

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6. Visual Design Philosophy

Inspired by the **Coinbase** design language, the CodexMap interface prioritizes **Trust and Clarity**:

- **High-Contrast Binary Palette:** Coinbase Blue (`#0052ff`) accents on a pure white/near-black grid.
- **Visual Hierarchy:** Ultra-tight display typography (1.00 line-height) and pill-shaped interactive CTAs.
- **Dynamic Depth:** Real-time micro-animations on node grades, signaling the system's "pulse."

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7. Strategic Roadmap

- **Phase 1:** Autonomous Debugging Agents (Fixing Reds without human intervention).

- **Phase 2:** CI/CD Guardrail Integration (Blocking merges on Drift > 0.40).
- **Phase 3:** Multi-Agent Observability (Tracking cross-agent communication protocols).

Built with pride by @Somu.ai for the OpenAI Codex Hackathon 2025.